ADVANCED DATA SCIENCE
Endterm Project (40 marks)

***Note**: Students should approach the project as a real-world client engagement, where they must understand the client's problem, define the decision to be supported, develop an evidence-based data science solution, and demonstrate how the proposed intervention can create measurable business value, such as revenue growth, cost reduction, improved customer retention, increased conversion, or operational efficiency. The project should provide evidence that the proposed intervention is likely to produce a measurable benefit for the client, rather than simply presenting a technically successful model.*

Instructions

1. Choose a Real-World Problem

Select a meaningful problem from business, healthcare, finance, marketing, operations, supply chain, or technology.

Start with the problem, not the dataset.

Clearly define:

- Problem
- Stakeholder
- Decision to be made
- Objective
- KPI / success metric
- Constraints

2. Define Hypotheses

Identify **2–3 questions/hypotheses** that your analysis will investigate.

Example:

> Customers with declining activity are more likely to churn.

3. Prepare and Understand the Data

Document the data source and variables. Perform appropriate:

ETL → Data Cleaning → Data Quality Checks → Integration → Analytics

Check missing values, duplicates, inconsistencies, outliers and possible bias.

4. Perform EDA

Use analysis and visualizations to generate **evidence**, not just graphs.

For important findings, explain:

Observation → Inference → Decision Relevance

5. Apply Appropriate Analytics

Use methods relevant to your problem, such as:

- Experimentation / A-B testing
- Causal inference
- Feature engineering
- Advanced ML
- Ensemble / boosting methods
- Hyperparameter optimization
- Recommendation / optimization

Do not force a method into the project simply because it appears in the syllabus.

6. Build and Evaluate the Solution

Compare appropriate approaches and justify the final model/solution.

Explain:

- Why the method was selected
- Why the evaluation metric was selected
- What the errors mean
- Whether the solution is reliable enough for the intended decision

7. Convert Results into a Decision

The final output must answer:

What should the stakeholder do based on the analysis?

Provide **5–10 actionable recommendations** supported by evidence.

8. Production and Responsible AI

Where relevant, demonstrate or discuss:

- Deployment / API

- MLflow / Docker
- Monitoring
- Data or concept drift
- Explainability using SHAP/LIME
- Fairness, bias, privacy and ethical issues

Use only technologies that are relevant to your problem.

9. Final Submission

Submit:

1. **Project Report**
2. **Code / Notebooks**
3. **Working Demo/Application**
4. **Presentation**

10. Final Evaluation: Decision Challenge

The evaluation will test **understanding, not just the final application**.

You may be asked:

- Why did you choose this problem?
- What decision does it support?
- Why did you choose this data?
- What did your analysis establish?
- Why this model?
- What are its limitations?
- What would you change if the business conditions changed?

A **surprise scenario** may be given, such as reduced budget, changed error costs, data drift, or fairness concerns. You must explain how your solution would respond.

Evaluation Focus

Component	**Focus**
Problem	Is the problem well defined?
Data	Is the data appropriate and reliable?
Analysis	Is there strong evidence?
Model	Is the approach justified?

Decision	Does it lead to an actionable decision?
Production	Can it work beyond the notebook?
Responsible AI	Is it explainable and responsible?
Defense	Do you genuinely understand your project?

Golden Rule: *Do not build a project just to demonstrate a model. Build a solution to support a decision.*

Project Evaluation: 40 Marks

Component	Marks	Evaluation Focus
Pre-submission	**10**	Problem definition, stakeholder, decision, objectives/KPIs, hypotheses, data source, and proposed methodology
Final Report	**10**	Data preparation, analysis, methodology, results, interpretation, limitations, and recommendations
Viva + Demo + Presentation	**20**	Understanding of the problem, technical implementation, working demo, interpretation, decision-making, and response to questions/surprise scenario
Total	**40**	